1 Forward kinematics models

In this section, we present the forward kinematics model we obtained for our CuRo6R robot using Denavit–Hartenberg parameters and also using Product of Exponentials. We also show numerical agreement between the two models.

1.1 DH model

The DH frames are assigned from the CuRo6R URDF, with frame $\{0\}$ at `base_link`, joint i rotating about z_{i-1}, and $q = 0$ matching the URDF home configuration. We use standard DH transforms:

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i). \tag{1}$$

Table 1 gives the resulting parameters and the geometric choices that determine them. The only fixed joint-angle offset is $\theta_3 - q_3 = \pi/2$. The full sequence of frame drawings is given in Appendix A.

Table 1: A non-unique solution for the DH parameters of CuRo6R. Lengths are in metres, angles are in radians.

i	d_i	θ_i	α_i	a_i	Geometric origin of the parameters
1	1	q_1	$-\pi/3$	0.8	The two base heights add: $d_1 = 0.5 + 0.5$; the lateral offset is $a_1 = 0.8$. Joint 2's axis is its local y axis, so the 30° URDF roll gives $\alpha_1 = 30° - 90° = -60°$.
2	1.25	q_2	$+\pi/2$	2	The joint-3 offset resolves into 1.25 along z_1 and 2 along x_2. At home $x_2 = x_1$, and the orthogonal joint axes give $\alpha_2 = +90°$.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Moving 0.75 along z_2 reaches the intersection with joint 4's axis: O_3 in step 3 of Appendix A, hence $a_3 = 0$. Choosing $x_3 = y_2$ gives the $+90°$ angle offset and $\alpha_3 = +90°$.
4	2.5	q_4	$-\pi/2$	0	The wrist centre lies $1.5 + 1 = 2.5$ along z_3 from O_3. Axes 4 and 5 intersect there; with $x_4 = x_3$ at home, $\alpha_4 = -90°$.
5	0	q_5	$+\pi/2$	0	Axes 5 and 6 meet at the wrist centre, so $O_5 = O_4$ and $d_5 = a_5 = 0$ [Appendix A, step 5]. Keeping $x_5 = x_4$ gives $\alpha_5 = +90°$.
6	1.5	q_6	0	0	The endpoint is 1.5 along z_5 from the wrist centre. Choosing $z_6 = z_5$ and $x_6 = x_5$ at home gives $a_6 = \alpha_6 = 0$.

Endpoint alignment. The endpoint E is `link_6`. Its origin coincides with O_6, while $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. The fixed alignment is therefore appended to the DH chain to recover the URDF endpoint pose:

$$^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{2}$$

1.2 PoE model

The product of exponentials (PoE) model uses the joint axes at the home configuration $q = 0$. For each joint, ω_i is its unit axis direction and r_i is its physical origin. All vectors are expressed in `base_link`; the endpoint is `link_6`, as in the DH model.

Axis directions. The rotation axes were identified visually in a URDF view at home. Their positive directions follow the URDF joint convention. The fixed tilt $\gamma = \pi/6$ gives the sine and cosine components of the tilted axes. In particular, $\omega_2 = R_x(\gamma)e_y$ and $\omega_3 = \omega_5 = R_x(\gamma)e_z$, where e_x, e_y, e_z are the base unit vectors.

Joint origins. Starting at joint 1, each origin was found by adding the relevant robot dimensions to the preceding origin. Each offset must first be expressed in the base frame. With lengths in metres, this gives

$$\begin{aligned} r_1 &= \tfrac{1}{2}e_z, & r_2 &= r_1 + \tfrac{4}{5}e_x + \tfrac{1}{2}e_z, \\ r_3 &= r_2 + 2e_x + \tfrac{5}{4}\omega_2, & r_4 &= r_3 + \tfrac{3}{2}e_x + \tfrac{3}{4}\omega_3, \\ r_5 &= r_4 + e_x, & r_6 &= r_5 + \tfrac{3}{2}e_x. \end{aligned} \tag{3}$$

For example, the offset from joint 2 to joint 3 combines a 2 m step along the base x axis and a 1.25 m step along joint 2's tilted axis. The resulting vectors are listed in Table 2, together with $v_i = -\omega_i \times r_i$.

Table 2: Home screw data. Each vector is written as a row; r_i uses the physical joint origin. Lengths are in metres.

i	ω_i^T	r_i^T	v_i^T
1	$(0, 0, 1)$	$(0, 0, \frac{1}{2})$	$(0, 0, 0)$
2	$(0, \frac{\sqrt{3}}{2}, \frac{1}{2})$	$(\frac{4}{5}, 0, 1)$	$(-\frac{\sqrt{3}}{2}, -\frac{2}{5}, \frac{2\sqrt{3}}{5})$
3	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{14}{5}, \frac{5\sqrt{3}}{8}, \frac{13}{8})$	$(\frac{7}{4}, -\frac{7\sqrt{3}}{5}, -\frac{7}{5})$
4	$(1, 0, 0)$	$(\frac{43}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$
5	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{53}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(\frac{7}{4}, -\frac{53\sqrt{3}}{20}, -\frac{53}{20})$
6	$(1, 0, 0)$	$(\frac{34}{5}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$

Forward model. Each pair (ω_i, v_i) defines a screw axis and its matrix form:

$$\xi_i = \begin{pmatrix} \omega_i \\ v_i \end{pmatrix}, \qquad \widehat{\xi}_i = \begin{pmatrix} [\omega_i]_\times & v_i \\ 0^T & 0 \end{pmatrix}, \qquad [\omega_i]_\times p = \omega_i \times p. \tag{4}$$

The endpoint has orientation $R_x(\gamma)$ and position r_6 at home. This gives its home pose M and the complete forward kinematics:

$$M = \begin{pmatrix} R_x(\gamma) & r_6 \\ 0^T & 1 \end{pmatrix}, \qquad \boxed{{}^0T_E(q) = e^{\widehat{\xi}_1 q_1} e^{\widehat{\xi}_2 q_2} \cdots e^{\widehat{\xi}_6 q_6} M.} \tag{5}$$

1.3 Verification

The DH and PoE models were compared at 514 configurations: home, each joint alone at $\pm\pi/2$, one mixed pose, and 500 uniformly sampled configurations in $[-\pi, \pi]^6$ (random seed 42). At each configuration, `check_dh_poe.m` computed the full pose Frobenius residual $\|\Delta T\|_F = \sqrt{\|\Delta p\|_2^2 + \|\Delta R\|_F^2}$, where Δ denotes the difference between the models, $\|\Delta p\|_2$ is the Euclidean position residual (positions in metres), and $\|\Delta R\|_F$ is the Frobenius rotation-matrix residual. The maximum full pose residual was below 10^{-14}, consistent with floating-point roundoff.

2 Jacobian and singularities

In this section we construct the geometric Jacobian of CuRo6R from the PoE model. The MATLAB PoE code keeps the intermediate transforms. These give each joint's current unit direction w_i and a point p_i on its axis, both in `base_link`. For a revolute joint, the Jacobian column is

$$ {}^0J_k(:, i) = \begin{pmatrix} w_i \times (p_k - p_i) \\ w_i \end{pmatrix}, \qquad i = 1, \ldots, 6. \tag{6} $$

The upper rows give linear velocity and the lower rows give angular velocity. We use $k = 6$ for the endpoint. Choosing $k = 5$ instead refers the same tool motion to the wrist centre.

Arm and wrist blocks. Here ${}^3J_5$ denotes this full Jacobian at the wrist centre, expressed along the physical link-3 axes (`link_3_d3`). Both velocity blocks are rotated by $({}^0R_3)^T$; the velocities remain relative to the base. All three wrist axes pass through p_5, so their linear columns vanish. With $c_i = \cos q_i$ and $s_i = \sin q_i$, the result has the form

$$ {}^3J_5 = \begin{pmatrix} J_{v,\mathrm{arm}} & 0 \\ J_{\omega,\mathrm{arm}} & J_{\omega,\mathrm{wrist}} \end{pmatrix}, \qquad J_{\omega,\mathrm{wrist}} = \begin{pmatrix} 1 & 0 & c_5 \\ 0 & -s_4 & c_4 s_5 \\ 0 & c_4 & s_4 s_5 \end{pmatrix}. \tag{7} $$

The subscripts v and ω denote linear and angular velocity. The 3×3 blocks $J_{v,\mathrm{arm}}$ and $J_{\omega,\mathrm{arm}}$ map the rates of joints 1–3 to these two velocities; $J_{\omega,\mathrm{wrist}}$ maps the rates of joints 4–6 to angular velocity. Their linear contribution at the wrist centre is zero.

We obtain an exact symbolic expression of ${}^3J_5$ in MATLAB, from which the determinant is extracted. The zero block of $J = {}^3J_5$ makes the determinant easy to evaluate:

$$ \det J(q) = -\sin q_5 \, \det J_{v,\mathrm{arm}}(q_2, q_3). \tag{8} $$

Using the DH lengths, define $U = a_2 + d_4 c_3$, $X = U c_2 + d_3 s_2$, and $Z = -U s_2 + d_3 c_2$. The explicit arm determinant is

$$ \det J_{v,\mathrm{arm}} = -\frac{d_4}{2} \left[a_1 Z c_3 - \sqrt{3}\, a_1 U s_3 - \sqrt{3}\, X (a_2 s_3 - d_2 c_3) \right]. \tag{9} $$

Changing the expression frame or velocity reference point preserves $\det J$, provided all six columns describe the same full tool motion. The wrist factor vanishes at $q_5 = 0$ or π modulo 2π. Axes 4 and 6 then lie on the same line, so one independent rotation is lost.

The arm factor depends jointly on q_2 and q_3. Thus the wrist and arm conditions separate, while the arm condition must be studied in the (q_2, q_3) plane, as shown in Figure 1.

A full configuration is regular exactly when both factors are nonzero. Numerically, we monitor the smallest singular value of J. It becomes zero at a singularity, and small values indicate that the Jacobian is close to losing rank.

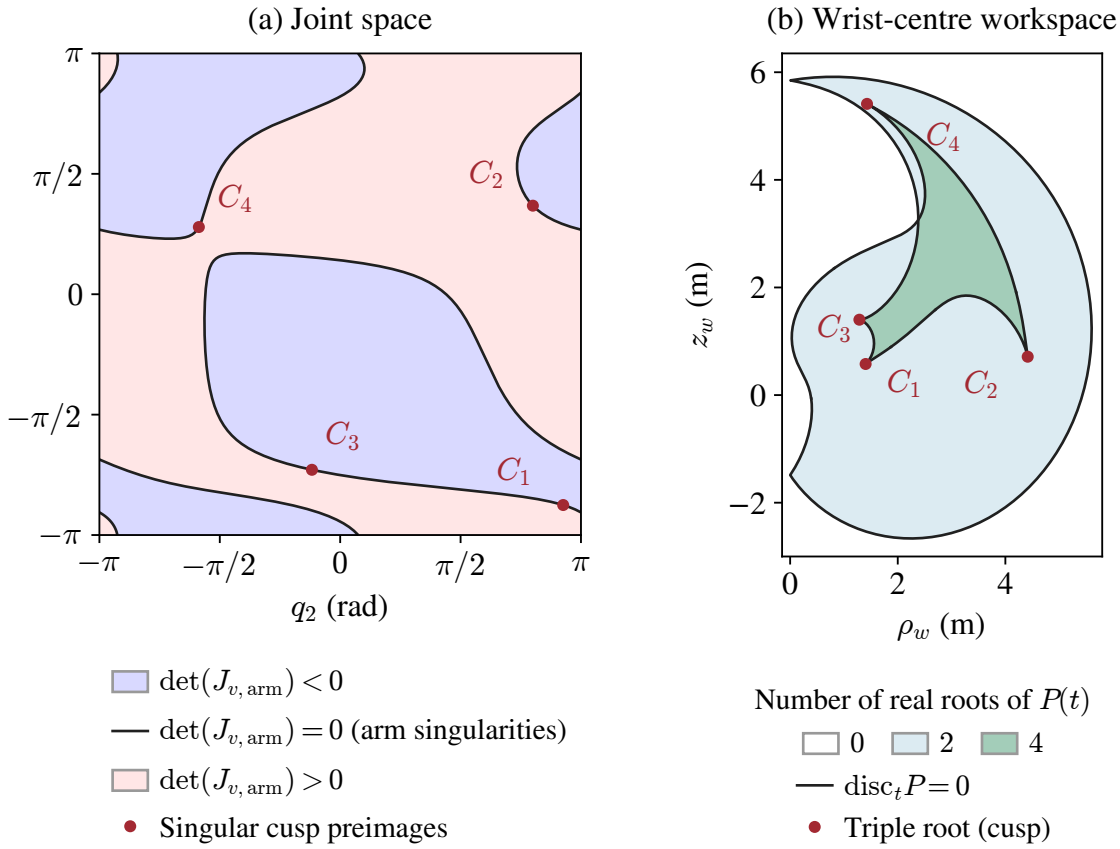


Figure 1: Arm singularities in (a) joint space and (b) the wrist-centre workspace, with $\rho_w = \sqrt{x_w^2 + y_w^2}$. In (a), colours give the sign of $\det J_{v,\text{arm}}$ and black curves show $\det J_{v,\text{arm}} = 0$. In (b), these curves map to real multiple roots of the IK quartic $P(t)$ in (23), derived in Subsection 3.4, where $\text{disc}_t\, P = 0$. Different shades of green give the number of distinct real roots away from the curves. For $\rho_w > 0$, this is the number of arm solutions before joint limits. Matching labels mark four cusps (triple roots) and their singular joint-space preimages. The existence of the cusps indicates that the CuRo6R's positioning chain is cuspidal, meaning it can change between IK solutions without crossing an arm singularity. Wrist singularities remain a separate condition, which can also avoided.

3 Analytical inverse kinematics

3.1 Frames, geometry and the wrist target

We seek the joint vectors $\mathbf{q} = (q_1, \ldots, q_6)^\mathsf{T}$ giving a prescribed pose of `link_6`. Frame 0 is `base_link`; frames $1, \ldots, 6$ are the URDF link frames. The q_i are URDF joint displacements from home.

Let C be the wrist centre, the origin of frame 5. The geometry is specified by

$$\begin{aligned} {}^0\mathbf{t}_2\big|_{q_1=0} &= \begin{pmatrix} a_1 \\ 0 \\ d_1 \end{pmatrix}, & {}^2\mathbf{t}_3 &= \begin{pmatrix} a_2 \\ d_2 \\ 0 \end{pmatrix}, \\ {}^3\mathbf{p}_C &= \begin{pmatrix} d_4 \\ 0 \\ d_3 \end{pmatrix}, \quad Q := {}^1R_2\big|_{q_2=0} = R_x(\alpha). \end{aligned} \tag{10}$$

Here d_1 combines the two vertical base offsets, and d_4 combines the x offsets from joint 3 to joint 4 and from joint 4 to joint 5. For the supplied robot, $(a_1, a_2, d_1, d_2, d_3, d_4, d_6) = (0.8, 2, 1, 1.25, 0.75, 2.5, 1.5)\,\mathrm{m}$ and $\alpha = \pi/6$. Write $c_i = \cos q_i$, $s_i = \sin q_i$, $c_\alpha = \cos\alpha$ and $s_\alpha = \sin\alpha$. The matrices R_x, R_y, R_z are right-handed rotations; $\mathbf{e}_x = (1,0,0)^\mathsf{T}$ and $\mathbf{e}_z = (0,0,1)^\mathsf{T}$ are coordinate unit vectors. The two-argument function $\mathrm{atan2}(s, c)$ denotes the four-quadrant inverse tangent, with the sine argument first.

All three wrist axes pass through C, so its position depends only on q_1, q_2, q_3. The origin of frame 6 lies a distance d_6 from C along its own x axis. The subscript d denotes a desired quantity, so the target pose gives the desired wrist position

$$\boxed{{}^0\mathbf{p}_{C,d} = {}^0\mathbf{t}_{6,d} - d_6\,{}^0R_{6,d}\mathbf{e}_x = \begin{pmatrix} x_d & y_d & z_d \end{pmatrix}^\mathsf{T}}. \tag{11}$$

3.2 Writing the wrist-centre position

Combining the relative translations and rotations gives us

$${}^0\mathbf{p}_C(\mathbf{q}) = \underbrace{d_1\mathbf{e}_z}_{{}^0\mathbf{p}_O} + \underbrace{R_z(q_1)}_{{}^0R_1} \underbrace{\Bigg[\underbrace{a_1\mathbf{e}_x}_{{}^1\mathbf{t}_2 - {}^1\mathbf{p}_O} + \underbrace{QR_y(q_2)}_{{}^1R_2} \underbrace{\Bigg(\underbrace{\begin{pmatrix} a_2 \\ d_2 \\ 0 \end{pmatrix}}_{{}^2\mathbf{t}_3} + \underbrace{R_z(q_3)}_{{}^2R_3} \underbrace{\begin{pmatrix} d_4 \\ 0 \\ d_3 \end{pmatrix}}_{{}^3\mathbf{p}_C} \Bigg)}_{{}^2\mathbf{p}_C} \Bigg]}_{{}^1\mathbf{p}_C - {}^1\mathbf{p}_O}. \tag{12}$$

This is also the position part of the product of exponentials. We evaluate it from the inside out. First, joint 3 gives us the shoulder-to-wrist coordinates in frame 2: ${}^2\mathbf{p}_C = (F_1, B, F_2)^\mathsf{T}$, where

$$F_1 := a_2 + d_4 c_3, \qquad B := d_2 + d_4 s_3, \qquad F_2 := d_3. \tag{13}$$

Next we apply joint 2, then the fixed tilt and the shoulder offset:

$$\begin{aligned} \underbrace{\begin{pmatrix} X \\ B \\ D \end{pmatrix}}_{\text{before the tilt}} &= R_y(q_2) \begin{pmatrix} F_1 \\ B \\ F_2 \end{pmatrix} = \begin{pmatrix} F_1 c_2 + F_2 s_2 \\ B \\ F_2 c_2 - F_1 s_2 \end{pmatrix}, \\ \underbrace{\begin{pmatrix} A \\ Y \\ H \end{pmatrix}}_{\text{before joint 1}} &= a_1\mathbf{e}_x + Q \begin{pmatrix} X \\ B \\ D \end{pmatrix} = \begin{pmatrix} a_1 + X \\ c_\alpha B - s_\alpha D \\ s_\alpha B + c_\alpha D \end{pmatrix}. \end{aligned} \tag{14}$$

A, Y, H are the wrist's two horizontal components and its height relative to the point on axis 1 at height d_1, in axes that rotate with joint 1. Applying the final base rotation gives us

$$x_d = Ac_1 - Ys_1, \qquad y_d = As_1 + Yc_1, \qquad z_d = d_1 + H. \tag{15}$$

3.3 Eliminating the first two joint angles

Let O be the point on axis 1 at shoulder height, ${}^0\mathbf{p}_O = (0, 0, d_1)^\mathsf{T}$. We now form three *known target quantities*:

$$h := z_d - d_1, \qquad \rho^2 := x_d^2 + y_d^2, \qquad \mathcal{R} := \rho^2 + h^2 = \|{}^0\mathbf{p}_{C,d} - {}^0\mathbf{p}_O\|^2. \tag{16}$$

h is the wrist height above O, ρ is wrist distance to axis 1, and $\mathcal{R}$ is the wrist squared distance to O. None changes when joint 1 rotates. For a solution, (15) gives $H = h$ and $\rho^2 = A^2 + Y^2$.

Rotation preserves norm. We apply that to the first relation in (14):

$$\underbrace{X^2 + B^2 + D^2}_{\|R_y(q_2)(F_1,B,F_2)^\mathsf{T}\|^2} = \underbrace{F_1^2 + B^2 + F_2^2}_{\|(F_1,B,F_2)^\mathsf{T}\|^2} \quad \Longrightarrow \quad X^2 + D^2 = F_1^2 + F_2^2. \tag{17}$$

The unchanged y component B cancels. Likewise, $(Y, H)^\mathsf{T} = \left(\begin{smallmatrix} c_\alpha & -s_\alpha \\ s_\alpha & c_\alpha \end{smallmatrix}\right)(B, D)^\mathsf{T}$ gives $Y^2 + h^2 = B^2 + D^2$. Hence

$$\begin{aligned} &\mathcal{R} = (a_1 + X)^2 + B^2 + D^2 = 2a_1X + F_3, \qquad h = s_\alpha B + c_\alpha D, \\ &F_3 := a_1^2 + F_1^2 + F_2^2 + B^2 = L + 2a_2d_4c_3 + 2d_2d_4s_3, \\ &L := a_1^2 + a_2^2 + d_2^2 + d_3^2 + d_4^2. \end{aligned} \tag{18}$$

The expansion of F_3 uses $c_3^2 + s_3^2 = 1$: F_3 depends only on q_3, while L is constant. With $\mathcal{R}, h$ known and a_1, c_α nonzero, a trial q_3 fixes

$$X = \frac{\mathcal{R} - F_3}{2a_1}, \qquad D = \frac{h - s_\alpha B}{c_\alpha}. \tag{19}$$

These required X, D can come from a real rotation q_2 only if their norm is correct. Substituting into (17) and multiplying by $4a_1^2$ gives us the *closure condition*:

$$\boxed{G(q_3) := (\mathcal{R} - F_3)^2 + \frac{4a_1^2}{c_\alpha^2}(h - s_\alpha B)^2 - 4a_1^2(F_1^2 + d_3^2) = 0.} \tag{20}$$

Only q_3 remains unknown. For $\alpha = 0$, $D = h$ and this becomes the lecture's untilted closure, with terminal arm length d_4.

3.4 Converting the closure into a quartic

We use the half-angle substitution

$$t = \tan(q_3/2), \qquad \Delta = 1 + t^2, \qquad c_3 = \frac{1 - t^2}{\Delta}, \qquad s_3 = \frac{2t}{\Delta}. \tag{21}$$

The hat below denotes a numerator after substitution. Multiplying by the common denominator Δ gives

$$\begin{aligned}\widehat{F}_1(t) &:= \Delta F_1 = a_2\Delta + d_4(1-t^2),\\ \widehat{B}(t) &:= \Delta B = d_2\Delta + 2d_4 t,\\ \widehat{F}_3(t) &:= \Delta F_3 = L\Delta + 2a_2 d_4(1-t^2) + 4d_2 d_4 t.\end{aligned} \tag{22}$$

Each numerator is quadratic in t. Multiplying (20) by $c_\alpha^2\Delta^2$ clears the denominators and yields a polynomial of degree at most four, because it contains squares of these quadratics:

$$\boxed{\begin{aligned}P(t) = c_\alpha^2(\mathcal{R}\Delta - \widehat{F}_3)^2 + 4a_1^2(h\Delta - s_\alpha\widehat{B})^2\\ - 4a_1^2c_\alpha^2(\widehat{F}_1^2 + d_3^2\Delta^2) = 0.\end{aligned}} \tag{23}$$

Every real root of the polynomial gives a $q_3 = 2\arctan t$. Since $\Delta > 0$ for real t, clearing denominators adds no real roots. We must test $q_3 = \pi$ separately in (20), because its half-angle tangent is undefined. This missing point is included in the maximum of four distinct q_3 candidates.

3.5 Recovering the remaining arm angles

For each candidate q_3, we evaluate (13) for F_1, F_2, B and use (19) for X, D. Combining their defining equations cancels s_2 in $F_1X + F_2D$ and c_2 in $F_2X - F_1D$, giving us

$$c_2 = \frac{F_1X + F_2D}{F_1^2 + F_2^2}, \qquad s_2 = \frac{F_2X - F_1D}{F_1^2 + F_2^2}, \qquad q_2 = \operatorname{atan2}(s_2, c_2). \tag{24}$$

The denominator is positive because $F_2 = d_3 \neq 0$, and the closure ensures $c_2^2 + s_2^2 = 1$. Next we set $A = a_1 + X$, $Y = c_\alpha B - s_\alpha D$ and recover the base angle from the horizontal equations:

$$q_1 = \operatorname{atan2}(\underbrace{Ay_d - Yx_d}_{\rho^2 s_1},\ \underbrace{Ax_d + Yy_d}_{\rho^2 c_1}), \qquad \rho > 0. \tag{25}$$

The common positive factor ρ^2 does not change the angle. If $\rho = 0$, the wrist centre lies on axis 1 and q_1 is free, in which case we choose a value in $[-\pi, \pi]$ and solve the wrist for that choice.

3.6 Recovering the wrist angles

For each arm solution, we undo its orientation using the transpose, then solve the remaining local x–z–x wrist rotations:

$$\begin{aligned}{}^0R_3 &= R_z(q_1)QR_y(q_2)R_z(q_3),\\ {}^3R_{6,d} &= ({}^0R_3)^{\mathsf{T}}\,{}^0R_{6,d} = R_x(q_4)R_z(q_5)R_x(q_6) =: (r_{ij}).\end{aligned} \tag{26}$$

Here r_{ij} is entry (i, j) of ${}^3R_{6,d}$. Multiplying the wrist rotations gives us five useful entries:

$$r_{11} = c_5, \quad r_{21} = c_4s_5, \quad r_{31} = s_4s_5, \quad r_{12} = -s_5c_6, \quad r_{13} = s_5s_6. \tag{27}$$

Let $\eta = \sqrt{r_{21}^2 + r_{31}^2} = |s_5|$. For $\eta > 0$, we choose each sign $\varepsilon \in \{-1, +1\}$, so that $s_5 = \varepsilon\eta$. Then

$$\boxed{\begin{aligned} &q_5 = \operatorname{atan2}(\varepsilon\eta, r_{11}), \\ &q_4 = \operatorname{atan2}(\varepsilon r_{31}, \varepsilon r_{21}), \qquad q_6 = \operatorname{atan2}(\varepsilon r_{13}, -\varepsilon r_{12}). \end{aligned}} \tag{28}$$

We must use the same sign in all three formulas. It selects one of two wrist configurations with the same orientation. If $\eta = 0$, joints 4 and 6 are coaxial, so only a sum or difference is fixed:

$$\begin{aligned} r_{11} = 1: &\quad q_5 = 0, \quad q_4 + q_6 = \operatorname{atan2}(r_{32}, r_{22}), \\ r_{11} = -1: &\quad q_5 = \pi, \quad q_6 - q_4 = \operatorname{atan2}(r_{32}, -r_{22}). \end{aligned} \tag{29}$$

We may set $q_4 = 0$ and recover q_6 from the appropriate relation.

3.7 Assembling and checking the solutions

For every real q_3 candidate, we recover the arm and both wrist branches. We represent angles in $[-\pi, \pi]$, enforce joint limits, remove duplicates modulo 2π, and check position and orientation by forward kinematics. We use numerical tolerances for roots, singularities and pose errors. Away from continuous singular families, at most four arm solutions each have two wrist branches, so there are **at most eight full-pose solutions**.

4 Connectivity analysis

A nonsingular change of solution returns the tool to its initial pose while leaving the robot in a different joint configuration and keeping the full pose regular throughout the motion. In this section, we show how a nonsingular change of solution can be achieved.

4.1 A straight arm path with fixed tool orientation

We first test a simple construction: connect two arm solutions by a straight segment in (q_1, q_2, q_3), then adjust the wrist to keep the tool orientation fixed. The tool position is free to move between the endpoints.

Selecting a candidate pair. For the initial configuration $\mathbf{q}^A$ below, the analytical solver returns eight solutions of $T_d = \mathrm{FK}(\mathbf{q}^A)$. Equation (8) gives two necessary conditions for a regular connection: $\det J_{v,\mathrm{arm}}$ and the wrist factor $\sin q_5$ must each retain their sign. Equal signs alone do not establish connectivity; the path between the endpoints must still be checked. After removing the initial solution and screening the signs and endpoint margins, one candidate remains, $\mathbf{q}^B$. In radians,

$$\begin{aligned} (\mathbf{q}^A)^\mathsf{T} &= (\ \ 0.1024,\ -1.1124,\ \ \ 1.8547,\ -1.4399,\ -0.3835,\ -2.6486), \\ (\mathbf{q}^B)^\mathsf{T} &\simeq (-2.2287,\ -2.1482,\ -0.2788,\ \ \ 0.6181,\ -0.3524,\ -1.7072). \end{aligned} \tag{30}$$

These are solutions 8^- and 4^- in Figure 2; the superscript denotes $\operatorname{sign}(\sin q_5)$.

From the arm segment to the full motion. Writing $\mathbf{q}_a = (q_1, q_2, q_3)^\mathsf{T}$, we prescribe

$$\mathbf{q}_a(s) = (1 - s)\mathbf{q}_a^A + s\mathbf{q}_a^B, \qquad 0 \le s \le 1, \tag{31}$$

where s is a path advancement parameter. At each sample, we compute $({}^0R_3(\mathbf{q}_a(s)))^{\mathsf{T}} R_d$ and recover q_4, q_5, q_6 using (28), always choosing the initial wrist sign. The angles are unwrapped towards the previous sample before testing $-\pi \le q_i \le \pi$; wrapping an angle back into this interval would hide a physical joint-limit crossing. With $\mathbf{p}_C$ the wrist centre, the resulting tool path is $\mathbf{p}_6(s) = \mathbf{p}_C(\mathbf{q}_a(s)) + d_6 R_d \mathbf{e}_x$. Both arm endpoints reach the same wrist centre, so $\mathbf{p}_6(1) = \mathbf{p}_6(0)$, although $\mathbf{q}(1) \neq \mathbf{q}(0)$. Thus a straight arm segment produces a closed, curved workspace path.

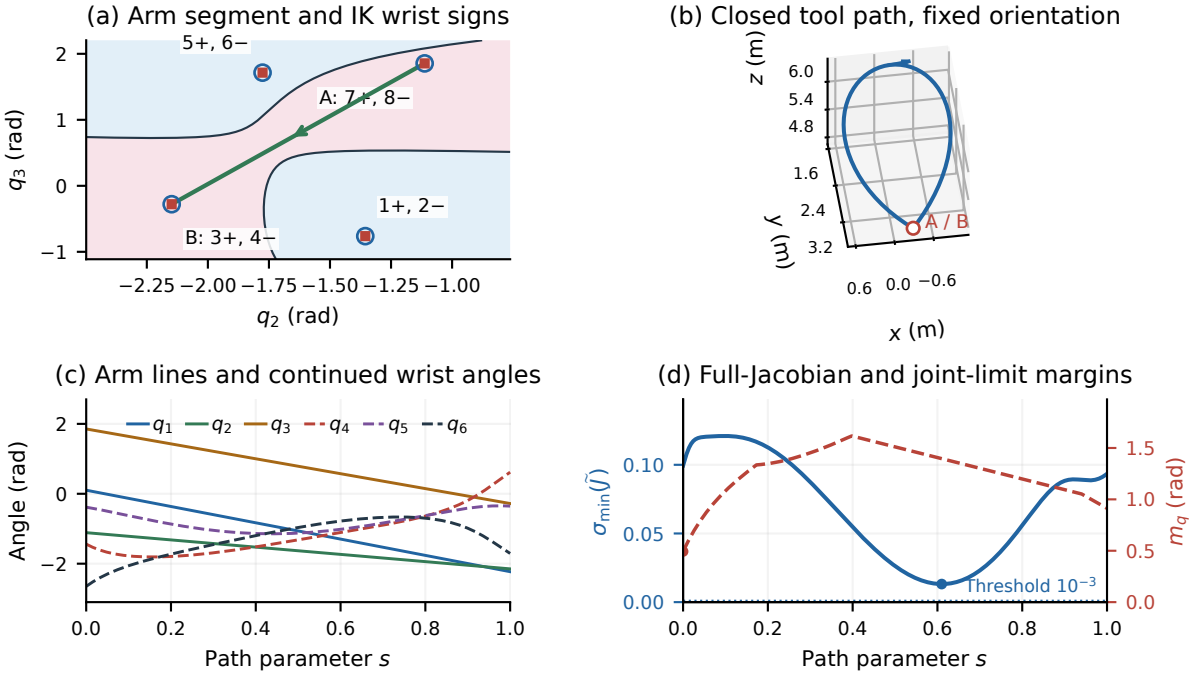


Figure 2: A fixed-orientation change from $A = 8^-$ to $B = 4^-$. In (a), black curves are $\det J_{v,\mathrm{arm}} = 0$; blue and pink denote negative and positive values, respectively. Each arm point carries both wrist signs, shown by a circle (+) and square (−). The green segment stays in one regular arm region. Panels (b)–(d) show the tool loop, all six joint paths, and the sampled margins.

Validation and interpretation. We evaluate 10 000 equally spaced samples, monitoring the arm sign, $|\sin q_5| > 10^{-6}$ and the full tool Jacobian. To compare its linear and angular rows, we use a characteristic length $L = 2\,\mathrm{m}$:

$$\tilde{J} = \mathrm{diag}(L^{-1} I_3, I_3)\, {}^0J_6, \qquad m_q(s) = \min_i \{q_i(s) + \pi,\ \pi - q_i(s)\}. \tag{32}$$

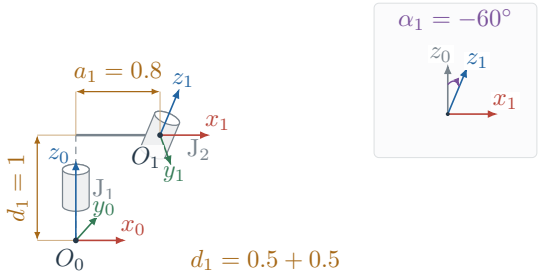
The minimum $\sigma_{\min}(\tilde{J})$ is 0.0132, above the 10^{-3} threshold; the minimum joint clearance is 0.493 rad and $\min |\sin q_5| = 0.337$. Position closure is 2.22×10^{-15} m and the maximum orientation error is 2.11×10^{-8} rad. Independent DH evaluation agrees with the PoE positions within 5.50×10^{-15} m. This numerically demonstrates a solution change for the selected pair. The checks concern sampled configurations, not every intervening point. A rejected segment would identify an arm, wrist or joint-limit obstruction along that attempt; it would not rule out a curved connection.

A DH frame constructions

Figure 3 collects all six frame-to-frame constructions at the home configuration ($q = 0$), using one fixed viewing direction and length scale. Lengths are in metres; angles are shown in degrees for readability.

1 Base to shoulder: $\{0\} \to \{1\}$

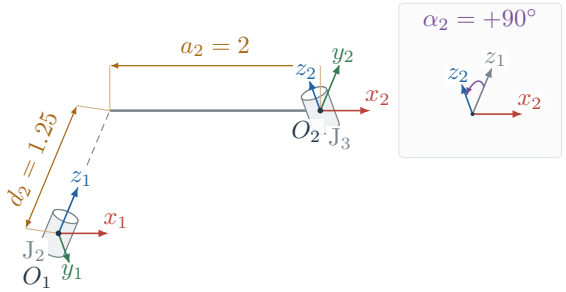
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$



The full base height is 1 m. The 30° URDF roll of joint 2 gives $\alpha_1 = 30° - 90° = -60°$.

2 Shoulder to elbow: $\{1\} \to \{2\}$

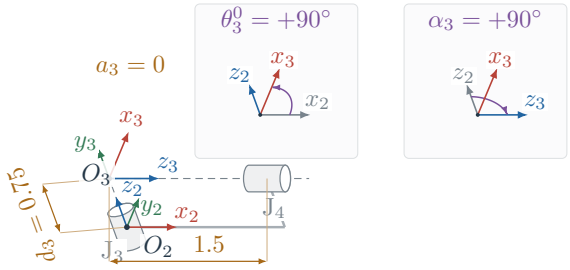
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$



The x axes remain parallel. The offset along z_1 and the common normal along x_2 locate O_2 on joint 3.

3 Axis intersection: $\{2\} \to \{3\}$

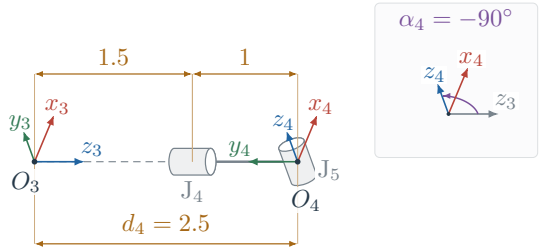
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$



O_3 is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed +90° offset in θ_3.

4 Reach the wrist: $\{3\} \to \{4\}$

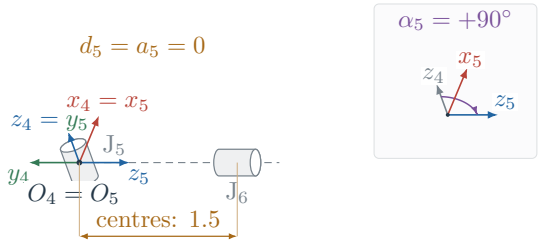
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$



The distance starts at O_3, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

5 Coincident origins: $\{4\} \to \{5\}$

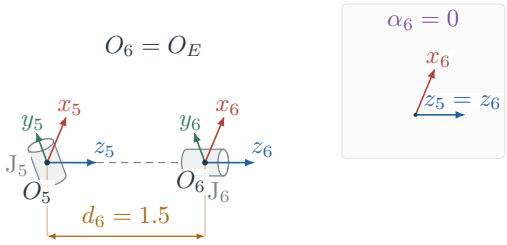
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$



The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

6 Wrist to endpoint: $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$



At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

Figure 3: Standard DH frame assignment. Cylinders denote physical revolute joints J_i; dots mark DH origins O_i. x, y and z axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Boxed angle sketches show the rotation $x_2 \to x_3$ about $+z_2$, giving θ_3^0, and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.